

BASIC ELECTRONICS

(24B11EC111)

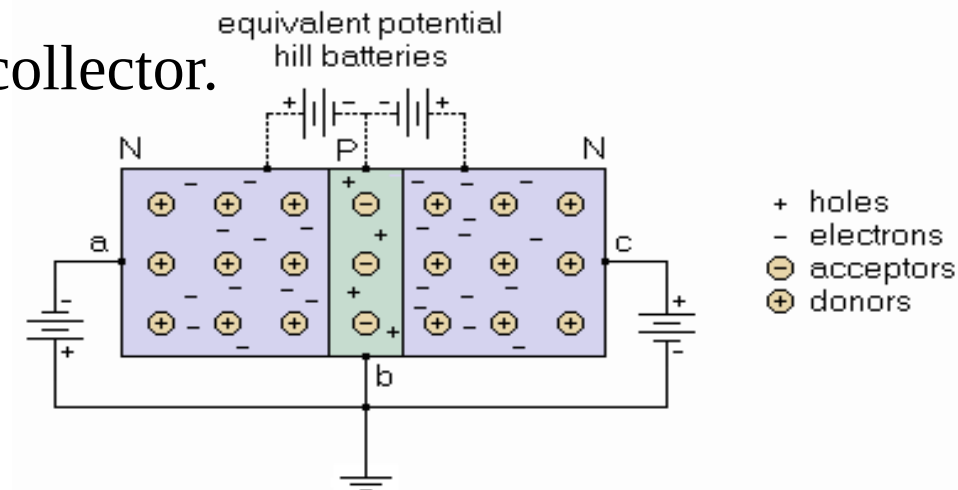
Lecture -2

Lecture Overview

- How transistor works
- Basic working
- Operation of npn transistor
- Operation of pnp transistor
- Transistor operating modes
- Transistor configurations and basic characteristics

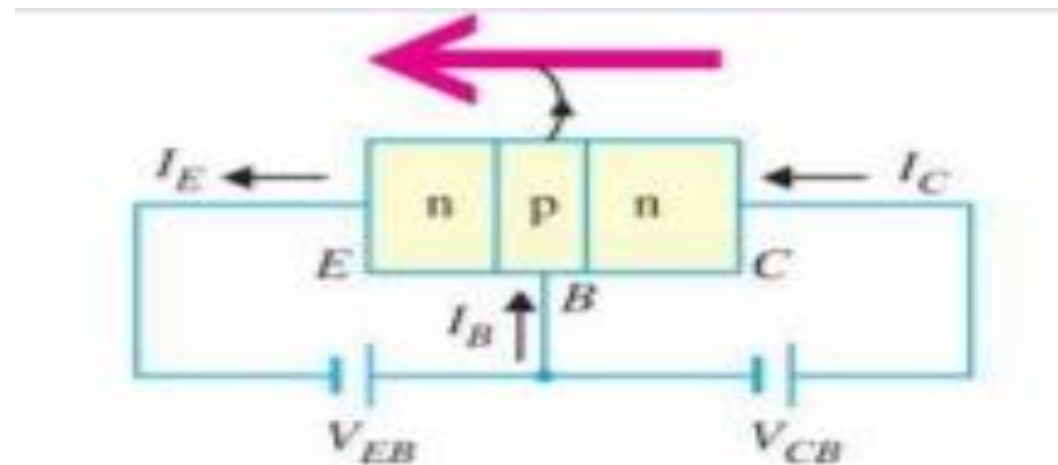
Basic Operation

- At the junction, a **built-in potential** is set up by the action of the **fixed donor and acceptor atoms**.
- Since the **emitter battery** acts to **flatten this emitter-base potential hill**, several electrons pass this barrier proportional to the value of emitter battery potential and enter the P base region.
- **Some of these electrons combine with holes in the P base region**, but most pass through and enter the N collector region.
- The loss of electrons in the **P base region remains low** (approximately five percent) because: (1) the **base section is thin**, and (2) the **potential at the collector-base junction acts to accelerate the electrons into the N collector region**.
- In the N region, the electrons are attracted to the positive collector.



Operation of npn Transistor

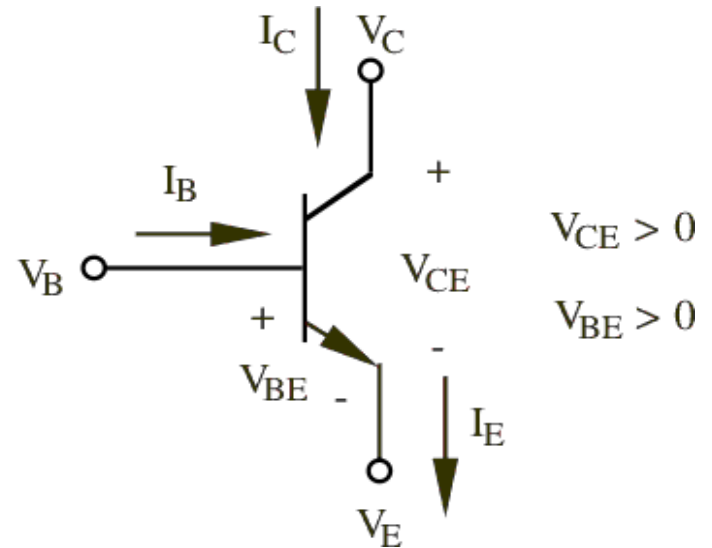
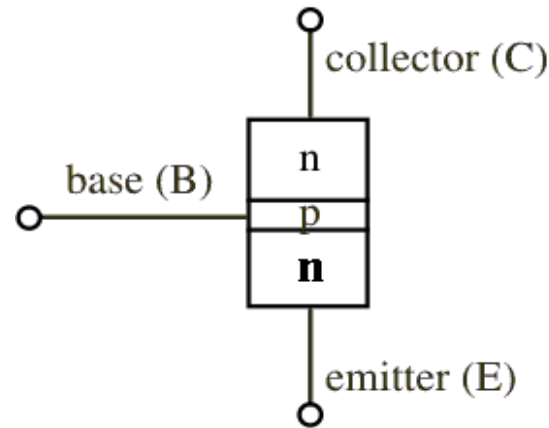
- **EB junction: Forward Biased**
 - The forward bias in the emitter–base junction causes electrons to move towards the base, constituting emitter current, I_e .
 - Electrons diffuse from Emitter to Base (from n to p).
- **CB junction: Reverse Biased**
 - There's a wide depletion layer on the Base-Collector junction due to reverse bias, which allows no flow of electrons from the collector to the base.



Operation of npn Transistor

- As the **electrons flow towards p-type base**, they try to recombine with holes. As the base is lightly doped, only a few electrons recombine with holes within the base.
- These recombined electrons constitute a small base current, I_b
- The remaining electrons cross the base and constitute collector current, I_c
- With $V_C > V_B > V_E$

$$\begin{aligned}I_E &= I_C + I_B \\V_{BE} &= V_B - V_E \\V_{CE} &= V_C - V_E \\I_C &= \beta I_B\end{aligned}$$

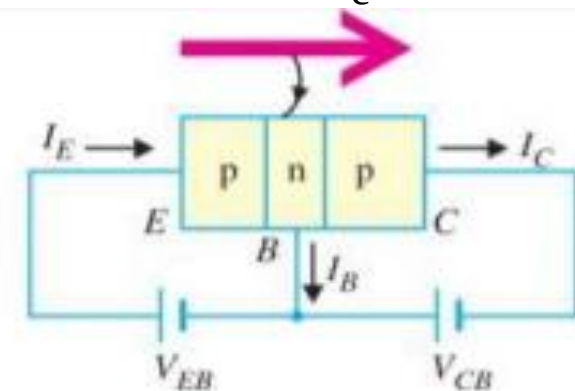


where β is the **current gain** of the transistor (typically 20–500).

That means a small base current controls a much larger collector current.

Operation of pnp Transistor

- **EB junction - Forward bias**
- The FB in EB junction causes holes to move towards the base, which constitutes emitter current, I_E
- **CB junction - Reverse bias**
- As the **holes flow towards the n-type base**, they try to recombine with electrons. As the base is lightly doped, **only a few holes recombine** with electrons within the base.
- These recombined holes constitute a **small base current, I_B**
- The remainder holes cross the base and constitute collector current, I_C



Transistor Operating Modes

➤ **Active Mode**

The base-emitter junction is forward-biased and the Collector-Base junction is reverse-biased.

➤ **Saturation Mode**

The base-emitter junction is forward-biased and the collector-base junction is also forward-biased. **(Always on)**

➤ **Cut-off Mode**

Both EB and CB junctions are reverse-biased. **(Always off)**

Transistor Configurations

- **Common Emitter:** current gain and voltage gain
- **Common Base:** no current gain but **voltage gain**
- **Common Collector:** current gain but no voltage gain

Transistor Basic Characteristics

- **Current Gain:**
 - α is the fraction of electrons that diffuse across the narrow **Base region**
 - $(1 - \alpha)$ is the fraction of electrons that **recombine with holes** in the Base region to create base current
- The **current Gain** is expressed in terms of the β (beta) of the transistor (often called h_{fe} by manufacturers).
- β (beta) is Temperature and Voltage dependent.

$$I_C = \alpha I_E$$

$$I_B = (1 - \alpha) I_E$$

$$\beta = \frac{I_C}{I_B} = \frac{\alpha}{1 - \alpha}$$

- **h** comes from the **hybrid-parameter model** (h-parameter model) used for small-signal analysis of transistors because the model mixes different units: resistance, conductance, and dimensionless gain.
- **F** = "forward" transfer characteristic (input $\rightarrow$ output).
- **E** = common-**emitter** configuration.
- So:
- **hFE** = forward current gain with emitter common = β .

Numerical

A transistor has a collector current of 10 mA and a base current of $40\text{ }\mu\text{A}$. What is the current gain of the transistor?

Numerical

A transistor has a collector current of 10 mA and a base current of 40 μA . What is the current gain of the transistor?

SOLUTION Divide the collector current by the base current to get:

$$\beta_{\text{dc}} = \frac{10 \text{ mA}}{40 \mu\text{A}} = 250$$

Numerical

A transistor has a current gain of 175. If the base current is 0.1 mA, what is the collector current?

Numerical

SOLUTION Multiply the current gain by the base current to get:

$$I_C = 175(0.1 \text{ mA}) = 17.5 \text{ mA}$$

Numerical

Determine the dc current gain β_{DC} and the emitter current I_E for a transistor where $I_B = 50 \mu A$ and $I_C = 3.65 \text{ mA}$.

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Determine the dc current gain β_{DC} and the emitter current I_E for a transistor where $I_B = 50 \mu\text{A}$ and $I_C = 3.65 \text{ mA}$.

Solution

$$\beta_{DC} = \frac{I_C}{I_B} = \frac{3.65 \text{ mA}}{50 \mu\text{A}} = 73$$

$$I_E = I_C + I_B = 3.65 \text{ mA} + 50 \mu\text{A} = 3.70 \text{ mA}$$

References

- [1] A. S. Sedra,, and K. C. Smith, ‘Microelectronic Circuits’. Oxford Oxford University Press, 7th Edition, 2012.
- [2] R. Boylestad and L. Nashelsky, ‘Electronic Devices and Circuit Theory’, PHI, 7e, 2001
- [3] D.C. Kulshreshtha, ‘Electronic Devices and Circuits’, New Age, 2e, 2006.